Numerical Road to Quantum Dots

I will use a 5-point stencil for the $\frac{d^2}{dx^2}$ operator:

Having:

- $f(x_k+h) = f(x_{k+1}) \approx f(x_k) + hf'(x_k) + \frac{h^2}{2}f''(x_k) + \cdots$
- $f(x_k-h) = f(x_{k-1}) \approx f(x_k) - hf'(x_k) + \frac{h^2}{2}f''(x_k) - \cdots$
- $f(x_k+2h) = f(x_{k+2}) \approx f(x_k) + 2hf'(x_k) + 2h^2f''(x_k) + \cdots$
- $f(x_k-2h) = f(x_{k-2}) \approx f(x_k) - 2hf'(x_k) + 2h^2f''(x_k) - \cdots$
- $f(x_k+3h) = f(x_{k+3}) \approx f(x_k) + 3hf'(x_k) + \frac{9}{2}h^2f''(x_k) + \cdots$
- $f(x_k-3h) = f(x_{k-3}) \approx f(x_k) - 3hf'(x_k) + \frac{9}{2}h^2f''(x_k) - \cdots$
- $f(x_k+4h) = f(x_{k+4}) \approx f(x_k) + 4hf'(x_k) + 8h^2f''(x_k) + \cdots$
- $f(x_k-4h) = f(x_{k-4}) \approx f(x_k) - 4hf'(x_k) + 8h^2f''(x_k) - \cdots$
- $f(x_k+5h) = f(x_{k+5}) \approx f(x_k) + 5hf'(x_k) + \frac{25}{2}h^2f''(x_k) + \cdots$
- $f(x_k-5h) = f(x_{k-5}) \approx f(x_k) - 5hf'(x_k) + \frac{25}{2}h^2f''(x_k) - \cdots$

Adding them to get rid of $f', f^{(3)}, f^{(4)}$:

$$\rightarrow -154f(x_{k+1}) + 214f(x_{k+2}) - 156f(x_{k+3}) + 61f(x_{k+4}) - 10f(x_{k+5}) = -45f(x_k) + (-77+428-702+488-125)h^2f''(x_k)$$

$$\rightarrow f''(x_k) = \frac{45f(x_k) - 154f(x_{k+1}) + 214f(x_{k+2}) - 156f(x_{k+3}) + 61f(x_{k+4}) - 10f(x_{k+5})}{12h^2}$$

$$\rightarrow 11f(x_{k-1}) + 6f(x_{k+1}) + 4f(x_{k+2}) - f(x_{k+3}) = 20f(x_k) + \left(\frac{11}{2}+3+8-\frac{9}{2}\right)h^2f''(x_k)$$

$$\rightarrow f''(x_k) = \frac{11f(x_{k-1}) - 20f(x_k) + 6f(x_{k+1}) + 4f(x_{k+2}) - f(x_{k+3})}{12h^2}$$

$$\rightarrow -f(x_{k-2}) + 16f(x_{k-1}) + 16f(x_{k+1}) - f(x_{k+2}) = 30f(x_k) + (-2+8+8-2)h^2f''(x_k)$$

$$\rightarrow f''(x_k) = \frac{-f(x_{k-2}) + 16f(x_{k-1}) - 30f(x_k) + 16f(x_{k+1}) - f(x_{k+2})}{12h^2}$$

$$\rightarrow -f(x_{k-3}) + 4f(x_{k-2}) + 6f(x_{k-1}) + 11f(x_{k+1}) = 20f(x_k) + \left(-\frac{9}{2}+8+3+\frac{11}{2}\right)h^2f''(x_k)$$

$$\rightarrow f''(x_k) = \frac{-f(x_{k-3}) + 4f(x_{k-2}) + 6f(x_{k-1}) - 20f(x_k) + 11f(x_{k+1})}{12h^2}$$

$$\rightarrow -10f(x_{k-5}) + 61f(x_{k-4}) - 156f(x_{k-3}) + 214f(x_{k-2}) - 154f(x_{k-1}) = -45f(x_k) + (-77+428-702+488-125)h^2f''(x_k)$$

$$\rightarrow f''(x_k) = \frac{-10f(x_{k-5}) + 61f(x_{k-4}) - 156f(x_{k-3}) + 214f(x_{k-2}) - 154f(x_{k-1}) + 45f(x_k)}{12h^2}$$

Summarized:

$$\ast\ f''(x_0) = \frac{45f(x_0)-154f(x_1)+214f(x_2)-156f(x_3)+61f(x_4)-10f(x_5)}{12h^2}$$

$$\ast\ f''(x_1) = \frac{11f(x_0)-20f(x_1)+6f(x_2)+4f(x_3)-f(x_4)}{12h^2}$$

$$\ast\ f''(x_k) = \frac{-f(x_{k-2})+16f(x_{k-1})-30f(x_k)+16f(x_{k+1})-f(x_{k+2})}{12h^2}$$

$$\ast\ f''(x_{n-1}) = \frac{-f(x_{n-3})+4f(x_{n-2})+6f(x_{n-1})-20f(x_n)+11f(x_{n+1})}{12h^2}$$

$$\ast\ f''(x_n) = \frac{-10f(x_{n-5})+61f(x_{n-4})-156f(x_{n-3})+214f(x_{n-2})-154f(x_{n-1})+45(x_n)}{12h^2}$$

Assuming Dirichlet boundary conditions: $\psi(x_{min}) = \psi(x_{max}) = 0$, we can just use the 4th-order central stencil

$$\ast\ \psi''(x_k) = \frac{-\psi(x_{k-2})+16\psi(x_{k-1})-30\psi(x_k)+16\psi(x_{k+1})-\psi(x_{k+2})}{12dx^2}$$

for all the points in the grid. This decision makes sure that our Hamiltonian is Hermitian $H=H^{\dagger}$

But notice for $k=1$ (first interior point):

$$\psi''(x_1) = \frac{-\psi_{-1}+16\psi_0-30\psi_1+16\psi_2-\psi_3}{12dx^2}$$

And two of these five values are not interior unknowns:

* $\underline{\psi_0 = \psi(x_0) = 0}$ from the imposed Dirichlet boundary condition.

* I originally made a mistake by assuming $\psi_{-1}=0$

sits one point outside the domain

* $\psi_{-1} = \psi(x_0-dx)$ → it is not a boundary values and is not zero. Let's find what it is

→ Notice that if we add the two Taylor expansions about x_0:

$$(1)\quad \psi(x_0+dx)+\psi(x_0-dx) = 2\sum_{j\geq 0}\frac{\psi^{(2j)}(x_0)}{(2j)!}(dx)^{2j}$$

Every odd derivative cancels in this sum and only the even ones survive. On the next page I show that the $j=0$ and $j=1$ terms both vanish so the sum starts at $j=2$ and

$$\psi(x_0-dx) \approx -\psi(x_0+dx)$$

$$\psi_{-1} \approx -\psi_1$$

* Showing which even derivatives vanish

→ Writing the time-independent Schrödinger equation as:

$$\psi'' = W\psi, \qquad W(x) = \frac{2m}{\hbar^2}(V(x) - E)$$

· for $j=0$: $\boxed{\psi(x_0)=0}$ → This is the Dirichlet condition itself.

· for $j=1$: $\boxed{\psi''(x_0) = W(x_0)\psi(x_0) = W(x_0)\cdot 0 = 0}$

· for $j=2$:

$$\rightarrow \psi''' = (W\psi)' = W'\psi + W\psi'$$

$$\rightarrow \psi'''' = W''\psi + 2W'\psi' + W\underbrace{\psi''}_{W\psi} = W''\cancel{\psi} + 2W'\psi' + W^2\cancel{\psi} = 2W'\psi$$

vanish at x_0

$$\Rightarrow \boxed{\psi''''(x_0) = 2W'(x_0)\psi'(x_0) = \frac{4m}{\hbar^2}V'(x_0)\psi'(x_0)}$$

So, truncating the sum from Eq. 1:

$$\psi(x_0+dx) + \psi(x_0-dx) = \frac{\psi''''(x_0)}{12}(dx)^4 + O(dx^6)$$

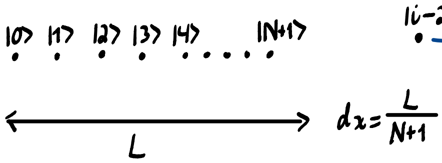
Thus,

(2) $\boxed{\psi_{-1} = -\psi_1 + \frac{\psi''''(x_0)}{12}dx^4 + O(dx^6) \approx -\psi_1}$

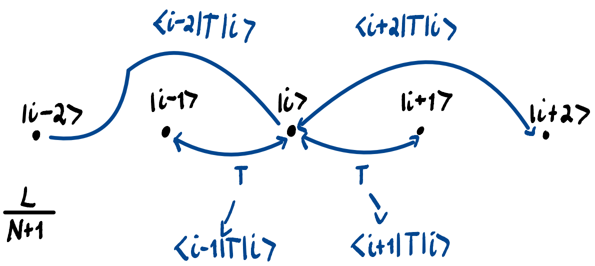
* Implemented in the code by adding $\frac{1}{12dx^2}$ to the first and last entry of the diagonal

→ 1-dimensional

$$H = T + V = \frac{p^2}{2m} + V = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V$$



$dx = \frac{L}{N+1}$



$$\Rightarrow \langle i'|H|i\rangle = \langle i'|T|i\rangle + \langle i'|V|i\rangle$$

$$= -\frac{\hbar^2}{2m}\left(\frac{-\langle i'|i-2\rangle + 16\langle i'|i-1\rangle - 30\langle i'|i\rangle + 16\langle i'|i+1\rangle - \langle i'|i+2\rangle}{12(dx)^2}\right) + V(x_i)\delta_{i'i}$$

$$= -\frac{\hbar^2}{24m(dx)^2}\left(-\delta_{i',i-2} + 16\,\delta_{i',i-1} - 30\,\delta_{i',i} + 16\,\delta_{i',i+1} - \delta_{i',i+2}\right) + V(x_i)\delta_{i'i}$$

Therefore,

$$\Rightarrow \psi''_i = \frac{-\psi_{i-2} + 16\psi_{i-1} - 30\psi_i + 16\psi_{i+1} - \psi_{i+2}}{12\,dx^2}$$

and substituting Eq. 2 for $i = 1, N$

$$\Rightarrow \psi''_1 = \frac{(-30+1)\psi_1 + 16\psi_2 - \psi_3}{12\,dx^2} \qquad \Rightarrow \psi''_N = \frac{(-30+1)\psi_N + 16\psi_{N-1} - \psi_{N-2}}{12\,dx^2}$$

I listed 9 potential functions:

1) Harmonic Oscillator

$$V(x) = \frac{1}{2} m\omega^2 x^2 \quad , \quad E_n = \hbar\omega\left(n + \frac{1}{2}\right) \quad , \quad n = 0, 1, 2, 3, 4, \ldots$$

2) Anharmonic Oscillator

$$V(x) = \frac{1}{2}\left(ax^2 + bx^4\right) \quad , \quad E_n \approx \hbar\omega\left(n + \frac{1}{2}\right) + \frac{3b\hbar^2}{8m^2\omega^2}\left(2n^2 + 2n + 1\right) \quad , \quad n = 0, 1, 2, 3, \ldots$$

* $a = m\omega^2$

Perturbation theory:

$$H' = \frac{1}{2} bx^4 \quad , \quad x = \sqrt{\frac{\hbar}{2m\omega}}\left(a^+ + a\right) \quad , \quad a^+|n\rangle = \sqrt{n+1}\,|n\rangle \quad , \quad a|n\rangle = \sqrt{n}\,|n-1\rangle$$

So:
$$(\hat{a}^+ + \hat{a})^4 = \hat{a}^{+4} + \hat{a}^{+3}\hat{a} + \hat{a}^{+2}\hat{a}\hat{a}^+ + \hat{a}^{+2}\hat{a}^2 + \hat{a}^+\hat{a}\hat{a}^{+2} + \hat{a}^+\hat{a}\hat{a}^+\hat{a} + \hat{a}^+\hat{a}^2\hat{a}^+ + \hat{a}^+\hat{a}^3 + \hat{a}\hat{a}^{+3} + \hat{a}\hat{a}^{+2}\hat{a} + \hat{a}\hat{a}^+\hat{a}\hat{a}^+ + \hat{a}\hat{a}^+\hat{a}^2 + \hat{a}^2\hat{a}^{+2} + \hat{a}^2\hat{a}^+\hat{a} + \hat{a}^3\hat{a}^+ + \hat{a}^4$$

From the expression found for $(\hat{a}^+ + \hat{a})^4$ we are just going to focus on the terms that contribute to $\langle n|\ldots|n\rangle$ since $\langle n|n+k\rangle = 0$, for that, the terms that we want need the same amount of $\hat{a}^+$ as $\hat{a}$, so:

$$\rightarrow \langle n|(\hat{a}^+ + \hat{a})^4|n\rangle = \langle n|\left(\hat{a}^{+2}\hat{a}^2 + \hat{a}^+\hat{a}\hat{a}^+\hat{a} + \hat{a}^+\hat{a}^2\hat{a}^+ + \hat{a}\hat{a}^{+2}\hat{a} + \hat{a}\hat{a}^+\hat{a}\hat{a}^+ + \hat{a}^2\hat{a}^{+2}\right)|n\rangle$$

$$= (\sqrt{n})(\sqrt{n-1})(\sqrt{n-1})(\sqrt{n}) + (\sqrt{n})(\sqrt{n})(\sqrt{n})(\sqrt{n}) + (\sqrt{n+1})(\sqrt{n+1})(\sqrt{n})(\sqrt{n}) + (\sqrt{n})(\sqrt{n})(\sqrt{n+1})(\sqrt{n+1}) + (\sqrt{n+1})(\sqrt{n+1})(\sqrt{n+1})(\sqrt{n+1}) + (\sqrt{n+1})(\sqrt{n+2})(\sqrt{n+2})(\sqrt{n+1})$$

$$= n(n-1) + n^2 + (n+1)n + n(n+1) + (n+1)^2 + (n+1)(n+2)$$

$$= n^2 - n + n^2 + n^2 + n + n^2 + n + n^2 + 2n + 1 + n^2 + 3n + 2$$

$$= 6n^2 + 6n + 3$$

$$\rightarrow \langle n|\frac{1}{2}bx^4|n\rangle = \frac{b\hbar^2}{8m^2\omega^2}\left(6n^2 + 6n + 3\right) = \frac{3b\hbar^2}{8m^2\omega^2}\left(2n^2 + 2n + 1\right)$$

3) Infinite Square Well

$$V(x) = \begin{cases} \infty, & x < 0,\ x > L \\ 0, & L \geq x \geq 0 \end{cases} \quad , \quad E_n = \frac{\hbar^2\pi^2}{2mL^2} n^2 \quad , \quad n = 1, 2, 3, \ldots$$

4) Finite Square Well

$$V(x) = \begin{cases} V_0, & x<0,\ x>L \\ 0, & L \geq x \geq 0 \end{cases}$$

- Solve for even states:

$$kL\tan\left(\frac{kL}{2}\right) = \kappa L$$

- Solve for odd states:

$$kL\cot\left(\frac{kL}{2}\right) = -\kappa L$$

with: $k = \frac{\sqrt{2mE}}{\hbar}$, $\kappa = \frac{\sqrt{2m(V_0-E)}}{\hbar}$

5) Linear Potential

$V(x) = Fx$, its solution reduces to an Airy equation and so the energy eigenvalues are determined by Airy zeros (considering Dirichlet boundary conditions

6) Soft Coulomb

$V(x) = -\frac{Z}{\sqrt{x^2+\varepsilon^2}}$, no closed form, it depends on ε

7) Quartic Single Well

$V(x) = V_0 x^4$, No analytic spectrum

8) Discrete Delta Well

$V(x) = -\alpha\delta(x)$, $E = -\frac{m\alpha^2}{2\hbar^2}$

9) Quartic Double Well

$V(x) = V_0(x^2-a^2)^2$, No closed form.

* Nearly degenerate ground pair
* Tunneling